IP1: scalar bilinears at one loop (engine A, exact)

$$O_{ab}=\int\omega\,(\iota_a A,\iota_b A),\quad K=\sum_I O_{e_I e_I},\quad u:=\frac{\hbar\,\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}_X\mathrm{ad}_Y)}{16\pi^2(X,Y)}=-\frac{g^2N}{32\pi^2}$$

$$S_{12}=(a_1,a_2)(b_1,b_2)+(a_1,b_2)(b_1,a_2),\qquad T_{12}=(a_1,b_1)(a_2,b_2),\qquad \tau=T_{12}/S_{12}$$

$$D_\varepsilon(r)=\frac{1-e^{-r^2/4\varepsilon}}{4\pi^2 r^2},\quad \hat D_\varepsilon(k)=\frac{e^{-\varepsilon k^2}}{k^2},\quad K_\varepsilon=-\Delta D_\varepsilon,\quad k^2\hat D_\varepsilon=e^{-\varepsilon k^2}$$

$$F_0\ni u\,\beta_2\int(dA|_{W^\perp},dA|_{W^\perp}),\qquad \psi=0,\qquad T=\infty,\qquad \ell=1$$

$$\langle O_1,O_2\rangle(x)=\frac{4\hbar\dim_{\mathfrak{g}}S_{12}}{16\pi^4|x|^4}\left[1+u\left(\tfrac{27}{2}+4\log\tfrac{3}{2}-8\beta_2\right)\right]+\frac{32\hbar\dim_{\mathfrak{g}}T_{12}}{16\pi^4|x|^4}\,u\left(\log\frac{|x|^2}{\ell^2}+\gamma_E-3\log 2\right)$$

$$(\mathrm{SE})=\tfrac{43}{2}+8\gamma_E-24\log 2+8L,\qquad (\mathrm{B})=-8\beta_2,\qquad L=\log|x|^2$$

$$(\mathrm{EX})=-8-4\gamma_E+8\log 2+4\log 3-4L+4\log\varepsilon$$

$$(\mathrm{QU})=(2\tau-1)(4L-4\log\varepsilon+4\gamma_E-12\log 2),\qquad (\mathrm{OC})=-8(1-\tau)\log\varepsilon$$

$$\sum\log\varepsilon=0,\qquad \sum L=8\tau,\qquad \sum\gamma_E=8\tau,\qquad \sum\log 2=-4-24\tau,\qquad \sum\log 3=4,\qquad \mathrm{const}=\tfrac{27}{2}$$

$$\Sigma(p)=u\left[\tfrac{9}{2\varepsilon}-2p^2\log\varepsilon-p^2\left(2\log p^2+2\gamma_E+2\log 2-\tfrac{43}{8}\right)\right]\quad(\text{coefficient of }(A(p),A(-p)))$$

$$\Sigma^{su(N)}/\tfrac{g^2N}{16\pi^2}:\ \mathrm{seagull}_V:-4\varepsilon^{-1};\ \mathrm{seagull}_\perp:-5\varepsilon^{-1};\ \mathrm{bubble}:\tfrac{1}{2\varepsilon}-2p^2(\log\varepsilon p^2+\gamma_E+\log 2)+\tfrac{13}{8}p^2$$

$$\mathrm{gaugino}:4\varepsilon^{-1}+4p^2(\log\varepsilon p^2+\gamma_E+\log 2)-7p^2;\qquad \Sigma_{ren}=\tfrac{g^2N}{16\pi^2}p^2(2\log p^2+2\gamma_E+2\log 2-\tfrac{43}{8})$$

$$M[0,0,2] = 1 - \gamma_E - \log 2 - \log(\varepsilon p^2), \quad M[0,0,3] = \tfrac{1}{4\varepsilon} + p^2\left(\tfrac{\log(\varepsilon p^2) + \gamma_E + \log 2}{6} - \tfrac{25}{72}\right)$$

$$M[1,1,4] = -\tfrac{\log(\varepsilon p^2) + \gamma_E + \log 2}{6} + \tfrac{13}{72}, \qquad c_{loc} = \int \tfrac{d^4k}{(2\pi)^4} \tfrac{e^{-2k^2}(e^{-k^2} - 1)}{k^4} = \tfrac{\log(2/3)}{16\pi^2}$$

$$J = \tfrac{D_0^2}{8\pi^2}\left(\log \tfrac{x^2}{\varepsilon} + \gamma_E - 3\log 2\right), \qquad R = \tfrac{D_0}{16\pi^2}\left(\log \tfrac{x^2}{\varepsilon} + 1 + \gamma_E - 3\log 2\right)$$

$$(k + k') \cdot (2q - k - k') = 2q^2 - k^2 - (q - k)^2 - k'^2 - (q - k')^2 + (k - k')^2$$

$$(\mathrm{EX}) = 32\pi^2 u\,[\,0 - 4(RD_0 + D_0^2 c_{loc}) + J + 2D_0^2 c_{loc}\,]/D_0^2, \qquad (\mathrm{QU}) = 32\pi^2 u(2\tau - 1)J/D_0^2$$

IP0 and Hermiticity

$$(O_{ab}(x))^{\Theta_I}=(x,x)^{-2}O_{\bar a\bar b}(I(x)),\qquad \big\langle a,b\big\rangle_{rad}=\big\langle a^{\Theta_I,\hbar},b\big\rangle$$

$$\big\langle O_{a_1b_1}(0),O_{a_2b_2}(0)\big\rangle_{rad}=\tfrac{4\hbar\dim\mathfrak{g}}{16\pi^4}\big[\bar S_{12}\big(1+u(\tfrac{27}{2}+4\log\tfrac{3}{2}-8\beta_2)\big)+8\bar T_{12}\,u(\gamma_E-3\log 2)\big]$$

$$\mathrm{Gram}_{tree}=\mathrm{Gram}_{tree}^{\dagger}>0:\quad \mathrm{diag}(2\ (\times 6),\ 1\ (\times 15))\cdot\tfrac{4\hbar\dim\mathfrak{g}}{16\pi^4}\ \text{on}\ e_Ie_J=\tfrac12\mathrm{Frobenius};\qquad \big\langle K,20'\big\rangle=0$$

$$\mathrm{Gram}_1=\mathrm{Gram}_1^{\dagger}\Leftrightarrow\beta_2\in\mathbb{R};\qquad \mathrm{Gram}_1-\mathrm{Gram}_1^{\dagger}=-16i\,\mathrm{Im}\beta_2\,u\,\mathrm{Gram}_{tree}/\hbar$$

$$J^{\Theta_n}=J,\ \ \psi(O^{\Theta_n})=\psi(O)^{\Theta_n}\ \Rightarrow\ \big\langle b,a\big\rangle_n=\overline{\big\langle a,b\big\rangle_n},\qquad \big\langle a,b\big\rangle_n:=\big\langle a^{\Theta_n},b\big\rangle$$

$$W(P,J')^{\Theta_n}=W(P,J'^{\Theta_n}),\quad J'^{\Theta_n}=J+\bar t_1a+\bar t_2b^{\Theta_n}\ \Rightarrow\ \big\langle a,b^{\Theta_n}\big\rangle=(-1)^{|a||b|}\overline{\big\langle a,b\big\rangle_n}$$

$$\big\langle\Lambda^{(1)}(Y)F,G\big\rangle=-\big\langle F,\Lambda^{(1)}(Y)G\big\rangle,\ Y=P_n-K_n\ \Rightarrow\ \big\langle a,b\big\rangle_{rad}=\big\langle e^{\frac{\pi}{4}\delta^h}a,e^{\frac{\pi}{4}\delta^h}b\big\rangle_n\ \Rightarrow\ \big\langle b,a\big\rangle_{rad}=\overline{\big\langle a,b\big\rangle_{rad}}$$

$$\mathrm{Gram}=\overline{\mathrm{Gram}}^{T},\qquad (X^{adj})^{adj}=X,\qquad \overline{X}^{T}\mathrm{Gram}=\mathrm{Gram}\,Y\Leftrightarrow\ \overline{Y}^{T}\mathrm{Gram}=\mathrm{Gram}\,X$$

$$\text{flow}:\quad \text{action}\to\text{vertices }V_3,V_4,\ \text{gaugino}\to\Sigma(p)\to\Sigma_{ren}\to\mathrm{FT}\to(\mathrm{SE})$$

$$\mathrm{EX}\to\text{numerator identity}\to\big\{2q^2I_5,\ K_\varepsilon\ \text{contacts}\big\}\to\text{localisation}\to\big\{J,R,c_{loc}\big\}\to(\mathrm{EX})$$

$$\mathrm{QU}\to\hat B(q)^2\to J\to(\mathrm{QU});\qquad \mathrm{OC}\to\partial_\delta I^{CT}(O)\to\text{trees}$$

$$\text{gates G1}-\text{G8}\to\text{engine A}=\text{engine B (per class)}\to\text{referee}\to\text{Thm}$$

Engine A gates

gate	pass
G1_selfenergy_singular_parts	True
G2_operator_counterterm	True
G3_log_eps_cancels	True
G4_log_x2_coefficient	True
G5_gammaE_cancels_on_20prime	True
G6_only_S12_T12	True
G7_IP0_gram_hermitian_positive	True
G8_colour	True

G1: self-energy singular parts (units)

graph	eps^-1	log eps	target eps^-1	target log eps
seagull_V	2	0	2	0
seagull_perp	5/2	0	5/2	0
bubble	-1/4	1	-1/4	1
gaugino	-2	-2	-2	-2
total	9/4	-1	9/4	-1

Engine A vs engine B (per class)

class	structure	coefficient	engine A (exact	engine A (num)	engine B	B digits	|A-B|	agree
T	S12	const	0	0	0		0.00e+00	yes
T	S12	log_x2	0	0	0		0.00e+00	yes
T	S12	log_eps	0	0	0		0.00e+00	yes
T	T12	const	0	0	0		0.00e+00	yes
T	T12	log_x2	0	0	0		0.00e+00	yes
T	T12	log_eps	0	0	0		0.00e+00	yes
SE	S12	const	-24*log(2) +	9.482192985	-24*log(2) +		0.00e+00	yes
SE	S12	log_x2	8	8	8		0.00e+00	yes
SE	S12	log_eps	0	0	0		0.00e+00	yes
SE	T12	const	0	0	0		0.00e+00	yes
SE	T12	log_x2	0	0	0		0.00e+00	yes
SE	T12	log_eps	0	0	0		0.00e+00	yes
B	S12	const	-8*beta_2	0	-8*beta_2		0.00e+00	yes
B	S12	log_x2	0	0	0		0.00e+00	yes
B	S12	log_eps	0	0	0		0.00e+00	yes
B	T12	const	0	0	0		0.00e+00	yes
B	T12	log_x2	0	0	0		0.00e+00	yes
B	T12	log_eps	0	0	0		0.00e+00	yes
EX	S12	const	-8 - 4*EulerGa	-0.369236060	-8 - 4*EulerGa		0.00e+00	yes
EX	S12	log_x2	-4	-4	-4		0.00e+00	yes
EX	S12	log_eps	4	4	4		0.00e+00	yes
EX	T12	const	0	0	0		0.00e+00	yes
EX	T12	log_x2	0	0	0		0.00e+00	yes
EX	T12	log_eps	0	0	0		0.00e+00	yes
QU	S12	const	-4*EulerGamm	6.008903507	-4*EulerGamm		0.00e+00	yes
QU	S12	log_x2	-4	-4	-4		0.00e+00	yes
QU	S12	log_eps	4	4	4		0.00e+00	yes
QU	T12	const	-24*log(2) +	-12.01780701	-24*log(2) +		0.00e+00	yes
QU	T12	log_x2	8	8	8		0.00e+00	yes
QU	T12	log_eps	-8	-8	-8		0.00e+00	yes
OC	S12	const	0	0	0		0.00e+00	yes
OC	S12	log_x2	0	0	0		0.00e+00	yes
OC	S12	log_eps	-8	-8	-8		0.00e+00	yes
OC	T12	const	0	0	0		0.00e+00	yes
OC	T12	log_x2	0	0	0		0.00e+00	yes
OC	T12	log_eps	8	8	8		0.00e+00	yes
F	S12	const	0	0	0		0.00e+00	yes
F	S12	log_x2	0	0	0		0.00e+00	yes
F	S12	log_eps	0	0	0		0.00e+00	yes
F	T12	const	0	0	0		0.00e+00	yes
F	T12	log_x2	0	0	0		0.00e+00	yes
F	T12	log_eps	0	0	0		0.00e+00	yes
TOTAL	S12	const	-8*beta_2 - 4	15.12186043	-8*beta_2 - 4		1.78e-15	yes

Referee: partial

id	found	claimed	agreement	severity
R1-J	{'a': '-1.00000000155',	{'a': -1, 'b': '-1.5022258	True	none
R2-R	{'a': '-1.00000000082',	{'a': -1, 'b': '-0.5022258	True	none
R3-cloc	{'momentum': '-0.0025(	-0.00256763779244922	True	none
R4-loc-i	{'rows': [['1e-2', '-0.001	-0.0025676377924492	True	none
R5-loc-ii	{'rows': [['1e-2', '-0.005	-0.0051352755848985	True	none
R6-G1	{'seagull_factors': [8.0,	{'Sigma_units_gsqN_16	True	none
R7-fermion-sign	{'chiral_trace_maxdev':	gaugino numerator 8 g^	True	none
R8-B	{'diff_rows': [['1e-2', '0.(	-0.27036284546148	True	none
R9-SigmaRen	-5.37499997977	-5.375 (= -43/8)	True	none
R10-pergraph	{'bubble_const': '-0.915	{'bubble': '-0.22128124	False	minor
R11-fourier	{'log q^2 @|x|=1.0': '-0	{'log q^2 @|x|=1.0': '-0	True	none
R12-G2	{'su2': {'validation': {'V	(e): O 2, (a,b)K -2; (f): O	True	none
R13-EX-QU	{'su2': {'tree_over_4dim	{'EX': '-N', 'QU': '-N(2T-S	True	none
R14-assembly	{'SE': '9.48219290485',	{'c_S': '27/2+4log(3/2) +	True	none
R15-gram-eigs	[1.0, 1.0, 1.0, 1.0, 1.0, 1	1 x20, 2 x1	False	minor
R16-contact	justified	contact term	True	none
R17-herm	valid	valid	True	minor
R18-hermrad	conclusion follows unde	valid	True	minor
R19-ip0-frame	consistent only if the fra	unconditional	None	minor
R20-spec-pw	eq:c4-pw has 2 (A(p),A(-	(A(p),A(-p))	False	minor
R21-unverified	[tp, lem:t-ft] remainder	None	None	none